

SHORT ANSWER TEST

CO1-ASSESSMENT TOOL - I

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COURSE CODE: CSA0702 - Computer Networks

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SLOT: B

DATE: 17/7/26

TOTAL MARKS:

20

OBTAINED MARKS:

19

Understanding of concept	Explanation	Application	Organisation	Accuracy	Clarity
25%	25%	20%	15%	10%	5%
5	5	4	3	1	1

25%
17/7/26

1) Network File Transfer Calculation

Given, File Size = 150 MB = 150,000,000 bytes

Segment size = 1500 bytes

Datalink Layer overhead = 40 bytes/frame

Bandwidth = 50 Mbps.

Sol,

$$1) \text{ Number of Segments} = \frac{\text{File size}}{\text{Segment size}} = \frac{150,000,000}{1500} = 100,000 \text{ segments}$$

$$2) \text{ Number of Frames} = 100,000 \text{ frames}$$

$$3) \text{ Data Link overhead} = 100,000 \times 40 = 4,000,000 \text{ bytes}$$

$$4) \text{ Transmission time} = 150 + 4 = 154 \text{ MB} = 1232 \times 10^6 \text{ bits}$$
$$= \frac{1232 \times 10^6}{50 \times 10^6} = \boxed{24.64 \text{ seconds}}$$

2) Sliding Window and Flow Control Calculation

Given, Sliding window size = 6 frames

User data per frame = 1024 bytes

Total frames transmitted = 48 frames

One frame lost in every window.

Sol,

$$1) \text{ Number of Transmission Windows} = \frac{\text{Total Frames}}{\text{Window Size}} \\ = \frac{48}{6}$$

$$\text{Number of Transmission Windows} = 8 \text{ Windows}$$

$$2) \text{ Total Retransmissions} = \text{One Lost Frame} \times \text{No. of Windows} \\ = 1 \times 8$$

$$\text{Total Retransmissions} = 8 \text{ retransmissions}$$

3) IP Packet Fragmentation Calculation

Given, IP Packet size = 12,000 bytes

MTU = 1500 bytes

IP Header per fragment = 20 bytes

Sol,

$$1) \text{ Maximum Data Per Fragment} = \text{MTU} - \text{IP Header} \\ = 1500 - 20$$

$$\text{Maximum Data Per Fragment} = 1480 \text{ bytes}$$

$$2) \text{ Number of Fragments} = \frac{\text{Packet Size}}{\text{Data per Fragment}} \\ = \frac{12000}{1480}$$

$$\text{Number of Fragments} = 8.11 \\ = 9 \text{ fragments}$$

Frames

$$\text{Total Header Overhead} = \text{Number of Fragments} \cdot \text{IP Header Size} \\ = 9 \times 20 = 180 \text{ bytes}$$

$$\boxed{\text{Total Header Overhead} = 180 \text{ bytes}}$$

4) Sliding Window and Flow Control Calculation:

Given, Sliding Window size = 12 frames

User data per frame = 1024 bytes

Total frames transmitted = 48 frames

One frame is lost in every window

Sol,

$$1) \text{ Number of Transmission Windows} = \frac{48}{12}$$

$$\boxed{\text{Number of Transmission Windows} = 4 \text{ windows}}$$

$$2) \text{ Total Retransmissions} = 1 \times 4 = 4$$

$$\boxed{\text{Total Retransmissions} = 4 \text{ retransmissions}}$$

5) Session Layer Synchronization Checkpoint Calculation

Given, Video Size = 600 MB

Synchronization Checkpoint after every 60 MB

Failure occurs after 390 MB

Sol,

$$1) \text{ Number of Checkpoints} = \frac{\text{Data transferred Before Failure}}{\text{Checkpoint Interval}}$$

$$\boxed{\text{Number of Checkpoints} = \frac{390}{60} = 6.5 = 6 \text{ Checkpoints}}$$

2) Data To be retransmitted = Data Transferred -
Failure - last ACK
= $390 - 6 \times 60$

Data To be retransmitted = 30 MB

6) Presentation Layer Compression and Encryption Calculation

Given, Original file size = 900 MB

Compression = 40%

Encryption overhead = 5% of compressed size.

Sol,

1) Compressed File = Original size $\times (1 - \text{Compression})$

Compressed File = $900 \times 1 - 0.40 = 540 \text{ MB}$

2) Encrypted File size = Compressed size $\times (1 + \text{Encryption overhead})$

Encrypted File size = $540 \times 1.05 = 567 \text{ MB}$

3) Percentage Reduction = $\frac{\text{Original size} - \text{Encrypted size}}{\text{Original size}}$

= $\frac{900 - 567}{900} \times 100$

= $\frac{333}{900} \times 100$

Percentage Reduction = 37%

TP - File Transfer Calculation:

Given, Data Transferred = 3 GB = 3000 MB

Effective throughput = 120 Mbps

Request size = 3 MB

Sol:

$$1) \text{ Number of Requests} = \frac{3000}{3} = 1000 \text{ requests}$$

$$2) \text{ Transmission Time} = \frac{3 \text{ GB}}{120 \times 10^6} \\ = \frac{24,000 \times 10^6 \text{ bits}}{120 \times 10^6}$$

$$\boxed{\text{Transmission Time} = 200 \text{ seconds}}$$

8) End-to-End Delay Calculation

Given, Application delay = 4 ms, Transport delay = 3 ms

Network delay = 5 ms, Data link delay = 2 ms

Data link = 2 ms, Physical = 1 ms

Propagation = 18 ms, Transmission = 12 ms

Sol,

$$\text{Processing delay} = 4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

$$\text{Total Delay} = 15 + 18 + 12$$

$$\boxed{\text{Total Delay} = 45 \text{ ms}}$$

9) Protocol overhead Calculation:

Given, Useful data = 80 MB

Frame size = 1600 bytes

Overhead = 80 bytes/frame

Sol,

$$1) \text{ Frames} = \frac{80,000,000}{1520} = 52,632$$

$$2) \text{ Overhead} = 52,632 \times 80 = 4,210,560 \text{ bytes}$$

$$3) \text{ Efficiency} = \frac{80,000,000}{84,210,560} \times 100 \approx 95\%$$

10) Multi-Layer OSI data Transmission Calculation

Given, File size = 240 MB

Overhead = 60 bytes/frame

Compression = 30%

Segment size = 1600 bytes

Sol,

$$1) \text{ Compressed size} = 240 \times 0.70 = 168 \text{ MB}$$

$$2) \text{ Segment} = \frac{168,000,000}{1600} = 105,000$$

$$3) \text{ Overhead} = 105,000 \times 60 = 6.3 \text{ MB}$$

$$4) \text{ Final data} = 168 + 6.3 = 174.3 \text{ MB}$$